

Separable Right-Support Nets Force UBCP

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when K has a countable π -basis, it asks for a noncommutative C^* -algebra characterization:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

Recall that a normed space has UBCP if its unit sphere is covered by countably many open balls whose radii are bounded and which all avoid a fixed neighborhood of the origin.

Definition 1. *Let A be a C^* -algebra. A sequence (e_n) of positive contractions in A is a separable right-support net if*

$$R_n := \overline{Ae_n}$$

is separable for every n , and

$$\|a\| = \sup_n \|ae_n\| \quad (a \in A).$$

Theorem 1 (right-support net criterion). *If a C^* -algebra A admits a separable right-support net, then A has UBCP.*

Idea of the Proof

The proof abstracts the useful part of the separable-essential-ideal argument. A positive contraction e gives a universal contraction $1 - 2e$ in the unitization. If $a \in S_A$ is almost normed by right multiplication with e , then the piece ae can be used as a countable center, provided the right ideal $\overline{Ae}$ is separable. The center is just a nearby dense point d , scaled to $2d$. The estimate

$$\|a - 2d\| \leq \|a(1 - 2e)\| + 2\|ae - d\|$$

gives a fixed radius, independent of a .

Proof

Proof of Theorem 1. For each n , choose a countable dense subset D_n of the closed unit ball of $R_n = \overline{Ae_n}$. Let

$$D_n^0 = \{d \in D_n : \|d\| > 13/16\}.$$

For every $d \in \bigcup_n D_n^0$, use the center $2d$ and the fixed radius $9/8$.

This is a countable family. Its radii are bounded, and every retained center has norm larger than $13/8$, so all balls avoid the open ball $B(0, 1/2)$. It remains to prove that these balls cover S_A .

Fix $a \in S_A$. Since (e_n) is norming on the right, choose n such that

$$\|ae_n\| > 7/8.$$

The element ae_n lies in the closed unit ball of R_n . Choose $d \in D_n$ with

$$\|d - ae_n\| < 1/16.$$

Then

$$\|d\| > 7/8 - 1/16 = 13/16,$$

so $d \in D_n^0$.

In the unitization of A , the element $1 - 2e_n$ is a contraction because e_n is a positive contraction. Hence

$$\begin{aligned} \|a - 2d\| &\leq \|a - 2ae_n\| + 2\|ae_n - d\| \\ &= \|a(1 - 2e_n)\| + 2\|ae_n - d\| \\ &< 1 + 1/8 = 9/8. \end{aligned}$$

Thus $a \in B(2d, 9/8)$. Since $a \in S_A$ was arbitrary, the displayed countable family is a UBCP cover. $\square$

Consequences

Corollary 1 (separable essential ideals). *If A contains a separable essential closed two-sided ideal, then A has UBCP.*

Proof. Let $I \triangleleft A$ be separable and essential. Choose a sequential positive contractive approximate identity (e_n) for I . Essentiality gives $\|ae_n\| \rightarrow \|a\|$ for every $a \in A$, and $\overline{Ae_n} \subset I$ is separable. Theorem 1 applies. $\square$

Corollary 2 (countable products). *Let (A_n) be a sequence of separable C^* -algebras. Then $\prod_n A_n$ has UBCP.*

Proof. Let e_n be the central projection onto the n -th coordinate. Then $\overline{(\prod_k A_k)e_n} \cong A_n$ is separable, and

$$\|a\| = \sup_n \|ae_n\| \quad (a \in \prod_k A_k).$$

Apply Theorem 1. □

Relation to the Pure-State Push

This packet does not prove a pure-state π -base classification. It clarifies what such a proof would need to produce. If a countable π -base in the pure state space could be converted into positive contractions e_n whose right ideals $\overline{Ae_n}$ are separable and which right-norm every element of A , then Theorem 1 would immediately give UBCP.

Pure-state excision is a plausible source of such e_n 's: it can create positive contractions whose high-level pure-state support lies in a prescribed weak-star neighborhood. The missing point is directional and quantitative: pure-state topology by itself has not yet been shown to force the separability of the corresponding right ideals, nor a countable set of local directions inside the excised corners.

Limitations

The theorem is a positive criterion, not an if-and-only-if classification. The hypothesis is stronger than the existence of a countable BCP blocker. In particular, an ideal generated by countably many elements inside a larger C^* -algebra can be nonseparable; a rank-one projection in $\mathcal{B}(H)$, for nonseparable H , generates $\mathcal{K}(H)$ as an ideal. The present theorem requires separability of the right-support pieces $\overline{Ae_n}$, not merely countable generation of an ideal.

Novelty and Search Bounds

Local indexes were searched for arXiv:2311.14257, BCP, UBCP, separable essential ideal, countable blocker, right ideal, right support, support net, and pure-state π -base. Web searches on June 17, 2026 for “separable right ideal uniform ball-covering property”, “right ideal ball-covering property C^* -algebra”, and “separable essential ideal ball-covering property C^* -algebra” did not locate this right-support-net criterion. The proof is an abstraction of the earlier lane-19 separable essential ideal argument, and should be reviewed as an agent-produced partial result.

Human Review Focus

Reviewers should check:

1. that the norming condition $\|a\| = \sup_n \|ae_n\|$ is used only to choose n with $\|ae_n\| > 7/8$;
2. that the dense set in the closed unit ball of $\overline{Ae_n}$ supplies a countable choice of centers;
3. that $\|1 - 2e_n\| \leq 1$ in the unitization for positive contractions e_n ;
4. the constants $7/8$, $1/16$, $13/16$, and $9/8$;
5. that the corollaries are genuinely special cases of the right support hypothesis.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Minzeng Liu, Rui Liu, Jimeng Lu, and Bentuo Zheng, *Ball covering property from commutative function spaces to non-commutative spaces of operators*, arXiv:2111.04921, 2021.
- [3] Bruce Blackadar, *Operator Algebras: Theory of C^* -Algebras and Von Neumann Algebras*, Encyclopaedia of Mathematical Sciences 122, Springer, 2006.
- [4] Run `fa_banach_001`, lane 19, *Separable Essential Ideals and the UBCP Wall*, solution packet `2311.14257_separable_essential_ideal_ubcp_reduction`, 2026.
- [5] Run `fa_banach_001`, lane 19, *BCP Forces a Faithful State and an Essential Blocker Ideal*, solution packet `2311.14257_bcp_forces_faithful_state_essential_blocker_ideal`, 2026.